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Zero-Shot Detection of Anti-CRISPR Proteins Using Protein Language Model Embeddings
Under Family-Level Holdout

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Abstract

Anti-CRISPR (Acr) proteins inhibit CRISPR–Cas immunity and are a crucial aspect of phage–host interactions, and are the foundations upon which genetic engineering advances are built. We asked whether protein language model (PLM) embeddings can generalize to novel Acr families absent from training data, simulating real unknown family discovery. Using ESM-2, we evaluated cosine similarity, k-nearest neighbors, and a linear logistic probe on mean-residue embeddings under a leave-one-family-out (LOFO) in-silico experiment design, with a 1:10 negative ratio. We compared ESM performance against homology search (BLASTp, phmmer) and simple k-mer baselines. ESM-2 (650M) significantly outperformed ESM-2 (3B) and all baselines, achieving micro-averaged AUROC ≈ 0.83 and AP ≈ 0.43 with a linear probe, while BLASTp and phmmer remained almost random under family holdout (AUROC $\approx 0.57/0.53$). UMAP charts showed somewhat separable Acr structure in embedding space. On a family-by-family basis, AUROCs revealed heterogeneity across families. Results indicate that PLM representations encode Acr-relevant features that transfer across families, offering a potential zero-shot filter for Acr discovery pipelines. We release code and aggregated results to facilitate reproducibility.

Introduction

Anti-CRISPR (Acr) proteins are small, rapidly evolving inhibitors that phages deploy to neutralize CRISPR–Cas immune systems in bacteria and archaea. By shutting down nuclease activity, Acrs shape phage–host arms races and are increasingly co-opted as precision “off-switches” for CRISPR technologies in biotechnology and therapeutics (Eitzinger et al., 2020). However, known Acrs are still few and far between. Lacking long conserved motifs and being potentially extremely short proteins, Acrs are encoded in incredibly diverse and disparate genes. These properties are repeatedly noted in curated Acr resources and discovery studies (e.g., AcrHub/AcrDB compendia and primary discovery reports). This creates a challenge for canonical homology search methods and also proves difficult for machine-learning algorithms which are hand-trained on limited Acr data. Hence, we address the problem by asking whether computational methods can recognize Acrs from unseen families.

Some of the first Acr predictors based on machine learning showed that detectable signatures are recognizable in models that are trained and tested within the same distribution. AcRanker, a tool developed by Eitzinger et al. in 2020, used hand-crafted sequence features with an XGBoost classifier and resulted in the experimental discovery of new inhibitors AcrIIA20 and AcrIIA21. PreAcrs expanded on this by combining multiple sequence and genomic features, reporting a stronger average performance across random splits (Zhu et al., 2022). At the same time that protein language models (PLMs) began to achieve widespread use, AcrNET combined ESM-1b embeddings with evolutionary and structural descriptors to further increase performance, and predict subclasses within Acr families (Li et al., 2023). Simultaneously, deep learning architectures like DeepAcr trained on raw sequences alone were shown capable of recovering previously discovered Acrs and predict plausible VI-B inhibitors (Wandera et al., 2022). These developments were enabled by community resources that curate Acr sequences, families, and genomic context, most importantly AcrHub and AcrDB, which standardize labels and gene-

neighborhood information and supply training and test material for computational work (Dong et al.; Zhang et al.).

Yet, many evaluations of these tools developed for Acr discovery don't sufficiently simulate the real discovery setting. The reason for this is that most prior models are supervised on known families and then evaluated using random holdouts from within the same family (Eitzinger et al., 2020; Zhu et al., 2022; Li et al., 2023). These methods of analysis speak to a model's capability to interpolate and find missing Acrs from within a previously established family, however say little about a model's capabilities when trying to uncover novel families. Curation studies emphasize that the Acr landscape is dominated by small families and unique candidates embedded in diverse genomic neighborhoods, which is the routine situation when scanning new phage genomes or metagenomes (Dong et al., 2021; Zhang et al., 2023). Under these settings, classical homology tools frequently underperform due to remote homologs falling below alignment sensitivity thresholds, and discriminative machine learning models run the risk of overfitting to family-specific details rather than to the tendency of a protein to be an Acr. Hence, the gap we aim to fill is to provide a family-level benchmark, evaluating whether a method can still identify members of a given Acr family if that family is held out during training.

Protein language models offer a promising route to filling this gap. Being trained at scale on a wide body of evolution-driven reasoning behind sequences, PLMs such as ESM-2 learn distributed representations that aim to encode biophysical regularities and functional constraints without explicit labels (Rives et al., 2021). This results in an embedder which attempts to generalize protein function from raw sequence. Across a wide range of tasks, PLM embeddings show zero-shot promise, where linear probes and simple similarity metrics are capable of retrieving functional signals that generalize beyond training distribution. Whether this zero-shot promise transfers to Acrs, which are short, varied, and frequently lineage-restricted, has not yet been systematically tested.

In this paper, we create a benchmark and provide an empirical answer. We build a leave-one-family-out (LOFO) evaluation in which every annotated Acr family is held out one at a time. The model is then tasked with identifying the held out family members in a mixture of 1:10 positives to negatives. We compare three readouts based on the ESM-2 embeddings (cosine similarity, k-nearest neighbors, and a balanced logistic probe) using two model sizes (650 M and 3 B parameters). As a baseline we take simple but informative methods, such as k-mer methods (Jaccard overlap and TF-IDF + logistic regression) and homology tools (BLASTp and phmmer) configured to exclude any hits to the held-out family, which prevents leakage. Subsequently, the embedding space is visualized with UMAP and micro-AUROC and AP are reported.

Across all metrics, the ESM-2 embeddings on the 650 million parameter model with a balanced logistic probe proves to be strongest, obtaining a micro AUROC of ≈ 0.83 and AP of ≈ 0.43 on LOFO at a 1:10 positives to negatives ratio. The homology baselines fail to predict left out Acr families, sitting at near-random performance, matching the intuition that these remote families are below alignment sensitivity. Although k-mer methods seem competitive on ROC, their precision-recall curves fall steeply at higher recall levels, accumulating false positives. We note two important observations: Firstly, in this imbalanced setting precision-recall is the most relevant

metric, as lab validation budget is the limiting factor in practical genome mining. Secondly, the smaller ESM-2 model, trained on 650 M parameters outperforms the larger model trained on 3 B parameters under identical folds and negatives. Hence, we presume the smaller model’s embedding geometry may be more linearly separable for Acr against non-Acr. This empirical note may guide future researchers in need of fast, effective, and lightweight screening.

In conclusion, we present Acr detection as a zero-shot generalization problem and demonstrate how PLM embeddings provide a clear and scalable. The aim of this paper is not to compete with other models trained on Acr discovery, but rather to expose the foundational baseline upon which many of these other models are built up on. This simple leakage-aware filter which generalizes across families can be combined with genomic context and structure based data for improved precision (Dong et al., 2021; Zhang et al., 2023). Our code and results are available upon request, and our goal is to standardize family-level evaluation and aid the community in the construction of models which may meet the true challenge of Acr discovery.

Research Materials and Approaches

Overview of evaluation design

To assess family-level generalization, we adopted a leave-one-family-out (LOFO) evaluation. In each fold, all members of one annotated Acr family are held out for testing, while all remaining families form the positive training pool. Negatives are drawn from specifically curated non-Acr proteins at a 1:10 positive:negative ratio. We then score test sequences using three readouts on PLM embeddings (cosine, k-NN density, and a class-balanced logistic probe) and compare against pre-established classical baselines (k-mer methods and homology search), reporting micro-averaged ROC and PR across folds. This group-wise split prevents leakage of family-specific signatures and directly tests extrapolation to unseen families, aiming to mirror real discovery. LOFO (a leave-one-group-out cross-validation) is standard when generalization across families rather than instances is the goal; it is functionally similar to remote-homology splits used in protein classification benchmarks

Curated positive set

We compiled a set of 341 anti-CRISPR (Acr) protein, across 76 annotated families, drawn from community resources AcrDB and AcrHub (Huang et al., 2020; Wang et al., 2020). Duplicates across these two databases were removed, and all proteins were saved in FASTA format. Family identifiers adhere to standardized naming conventions (e.g. AcrIIA1, AcrIIC2), hence sequences with the same label represent homologous inhibitors. The final set ranges from short peptides to a few hundred residues, which reflects typical Acr lengths.

Construction of realistic decoys (negatives)

We used real non-Acr proteins rather than random strings, because random sequences are not representative of the proteomes encountered in Acr discovery and make classification artificially easy. With the goal of approximating the sequences encountered during real experimental discovery, we curated negatives from three biologically grounded sources, filtering them to avoid trivial separation from the positive set.

1. Bacterial proteins, matched in length. We sampled reviewed Swiss-Prot entries from Bacteria with lengths 40–200 amino acids while excluding obvious CRISPR terms such as, “anti-CRISPR,” “acr,” “crispr”. This group matches Acrs in size and cellular context while intentionally excluding CRISPR machinery.
2. CRISPR-adjacent proteins. We included reviewed bacterial proteins annotated with Cas and related keywords (e.g., cas, csy, csm, cmr), 40–400 amino acids in length. These negatives were intended to be more challenging, as they occur in the same genomic neighborhoods as Acrs and can share weak compositional signals.
3. Phage small proteins. We drew Caudoviricetes proteins in the 40–200 amino acid range while excluding Acr keywords. This adds phage-encoded small ORFs that resemble Acrs in length and origin but are not known inhibitors.

We merged these different sources, deduplicated across this new dataset, and removed close homologs of positives using a permissive BLAST screen (dropping sequences with >30% pairwise identity over >50% coverage to any Acr). This conservative filter reduces the chance that the downstream methods succeed by detecting near-duplicates. For the primary experiments we formed 1:1 and 1:10 matched sets, with 341 and 3,410 negatives respectively.

Protein embeddings (ESM-2)

We embedded every protein sequence with ESM-2 at two model sizes, a 650M-parameter model and a 3B-parameter model. Each sequence was tokenized with the model’s alphabet, fed through the network, and we mean-pooled the final hidden layer across residues to obtain one fixed-length vector per protein. We then L2-normalized all vectors so that cosine similarity equals the dot product. Inference ran in half precision on a single CUDA GPU with small batch sizes to avoid memory pressure. Embeddings and metadata (protein identifier, family label, sequence length) were stored in columnar Parquet format for efficient downstream loading. The choice of ESM-2 follows prior observations that large protein language models learn general, label-free structure–function features from sequence corpora (Rives et al., 2021).

Leave-one-family-out (LOFO) evaluation

To test generalization to unseen families, we designed a leave-one-family-out protocol. For each Acr family in turn:

1. Hold out all sequences of that family for testing.
2. Training positives are all remaining Acr families.
3. We create a single global split of the negatives into train and test halves and reuse this split across all families. This avoids information leakage between folds.
4. The test set for the fold consists of all held-out family members mixed with a 1:10 sample of negatives drawn from the negative test half. The training set for supervised probes consists of training positives and the negative train half only.
5. We repeat this process over families and, where stochastic subsampling is involved, over a few seeds to stabilize estimates. We report micro-averaged metrics across all folds, and macro summaries over families with at least a minimal number of positives (e.g., ≥ 3).

This design simulates the real discovery setting in which a new Acr often belongs to a family not seen during training.

Readouts on embeddings

We evaluated three simple, label-efficient readouts over the ESM-2 embeddings:

- Cosine max-similarity. For each test protein, we compute cosine similarity to every training positive and use the maximum as the score. This is a zero-parameter, nearest-prototype view of the embedding space.
- k-nearest neighbors ($k=5$). We find the five nearest training positives in Euclidean distance on L2-normalized vectors and use the inverse of the mean distance as the score. This results in a smoothed max-similarity signal and can be more robust when training positives are sparse.
- Logistic regression (linear probe). We train an L2-regularized logistic classifier on training positives versus training negatives, with `class_weight="balanced"` and standardized inputs. This probe tests whether a linear separator can be learned between Acrs and non-Acrs in the embedding space without using any examples from the held-out family. Regularization strength was selected in a narrow, stable range (e.g., $C \approx 0.5-1.0$) and not tuned per family.

These readouts are purposefully simple, so that any improvements reflect the quality of the embeddings rather than a heavyweight classifier.

Baseline methods without embeddings

We compared the ESM 650M model against sequence-only baselines that do not use language-model features:

- k-mer Jaccard (k=3). This method involves splitting each sequence into (overlapping) 3mers and calculating the maximum Jaccard overlap when compared to any training positive. This proves exceptionally useful for short, conserved motifs.
- k-mer TF-IDF + logistic regression (k=3 and k=4). Achieved by breaking sequences into character k-mers, computing TF-IDF features, and training a balanced logistic classifier on training positives and training negatives. The vectorizer is fit only on the training split to avoid IDF leakage. The k=3 setting emphasizes short motifs; k=4 increases specificity and typically reduces chance matches.
- BLASTp. We build a protein database from training positives only. For each test query we collect hits excluding subjects from the held-out family and score the query by the best continuous alignment score, defined as $\text{bitscore} \times (\text{alignment length} / \text{query length})$. This rewards both similarity and coverage and yields a monotonic ranking for ROC/PR. We use permissive thresholds so weak, distant matches are retained.
- phmmer (HMMER3). We mirror the BLAST setup using phmmer against the training-positive database and take each query's best bit score while excluding hits to the held-out family.

These baselines aim to encapsulate the simple sequence similarities visible by simple algorithms, and highlight the added nuance of using a PLM.

Metrics and aggregation

Discovery is class-imbalanced, so we report both receiver operating characteristic (ROC) and precision–recall (PR) views:

- Micro-averaged AUROC and AP. We aggregate scores and labels across all held-out folds to produce a single ROC and PR curve per method, then compute AUROC and average precision (AP). Micro aggregation reflects the decision quality on the full test pool a practitioner would face.
- Macro summaries with confidence intervals. Where noted, we average AUROC or AP across families that have sufficient test positives (for example, ≥ 3) and report 95%

confidence intervals from repeated negative subsampling. This highlights variability across families.

We favor precision–recall when discussing practical screening, since laboratory validation budgets are limited and precision at moderate recall is often more actionable than a high ROC at low false-positive rates.

Visualization of embedding geometry

To visualize how Acrs and negatives are arranged in the embedding space, we project vectors to two dimensions using UMAP. For stability we first reduce to 50 principal components and then apply UMAP with typical parameters (`n_neighbors` 15–25, `min_dist` 0.08–0.10), primarily for speed and denoising; this is standard for protein-scale embedding visualizations. COSINE relationships are preserved well under modest PCA and the 50-D step removes tiny axes of variance that otherwise make UMAP/t-SNE unstable across runs. We plot two views: (i) Acrs versus a lightly sampled set of negatives, and (ii) Acrs colored by the largest families with negatives in the background. These plots are descriptive rather than evaluative and help relate quantitative performance to qualitative structure.

Reproducibility and implementation details

All experiments used fixed random seeds for subsampling and negative splits. We created a single disjoint train/test split of the negatives and reused it across all families and across model sizes so that comparisons are “apples to apples.” Embeddings were written to Parquet with identifiers, family labels, and sequence lengths to support traceability.- Evaluation code produces per-family CSV files and figure images in a structured directory so that results can be regenerated from raw sequences and configuration alone.

Results

Zero-shot detection of unseen Acr families with ESM-2 embeddings

We evaluated anti-CRISPR detection under a stringent leave-one-family-out (LOFO) protocol with a 1:10 positive-to-negative class ratio. In each fold, one Acr family was held out entirely for testing; training used the remaining families plus a disjoint training half of the negatives, and testing mixed the held-out family with the test half of the negatives. On this benchmark, the ESM-2 (650M) embedding + logistic probe achieved micro-AUROC ≈ 0.83 and micro-AP ≈ 0.43 , outperforming both simpler embedding readouts (cosine, kNN) and classical sequence baselines (Fig. 1A–B).

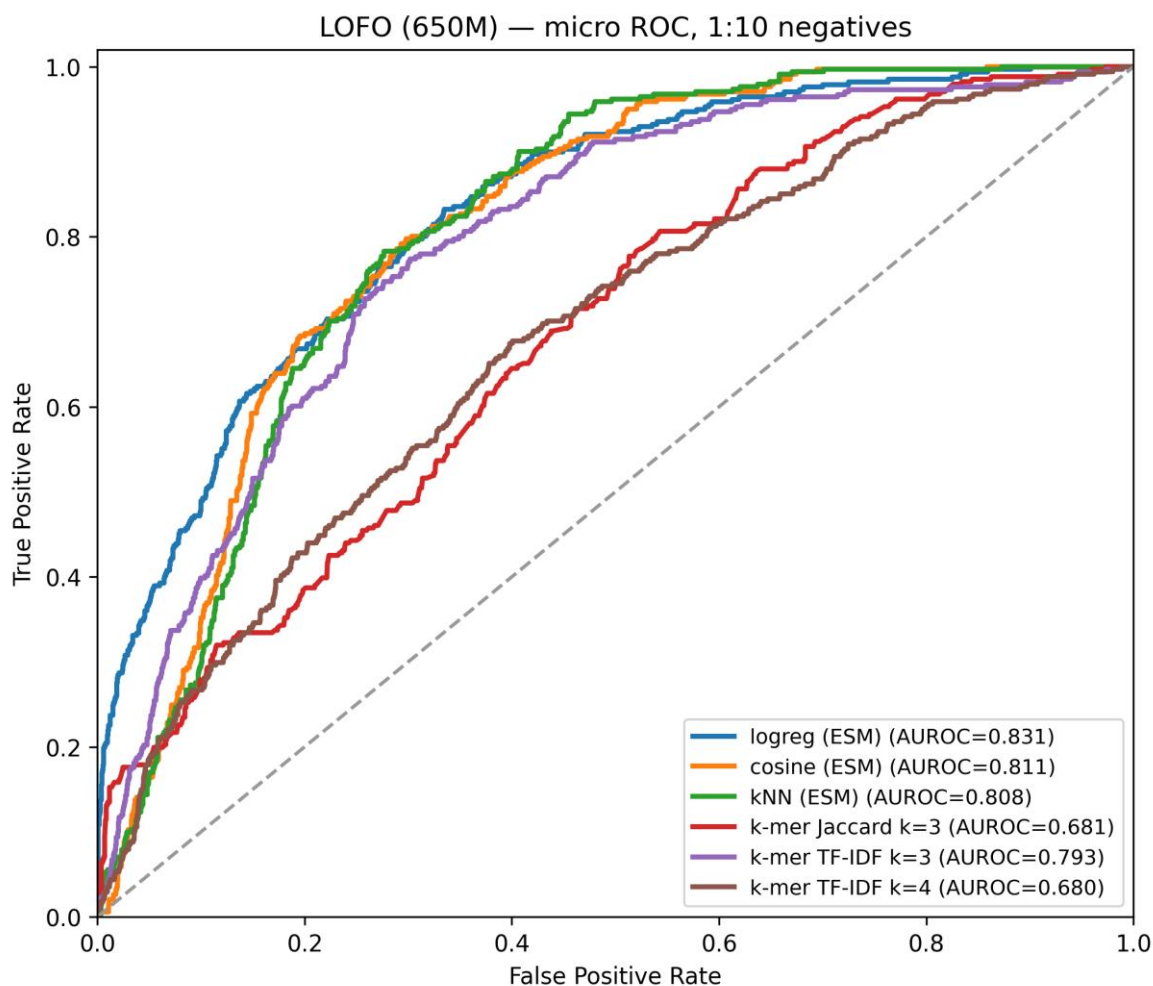


Figure 1A: Micro-ROC across methods. The ESM-2 (650M) embedding with a balanced logistic probe outperforms cosine, kNN, k-mer TF-IDF ($k=3/4$), and homology baselines.

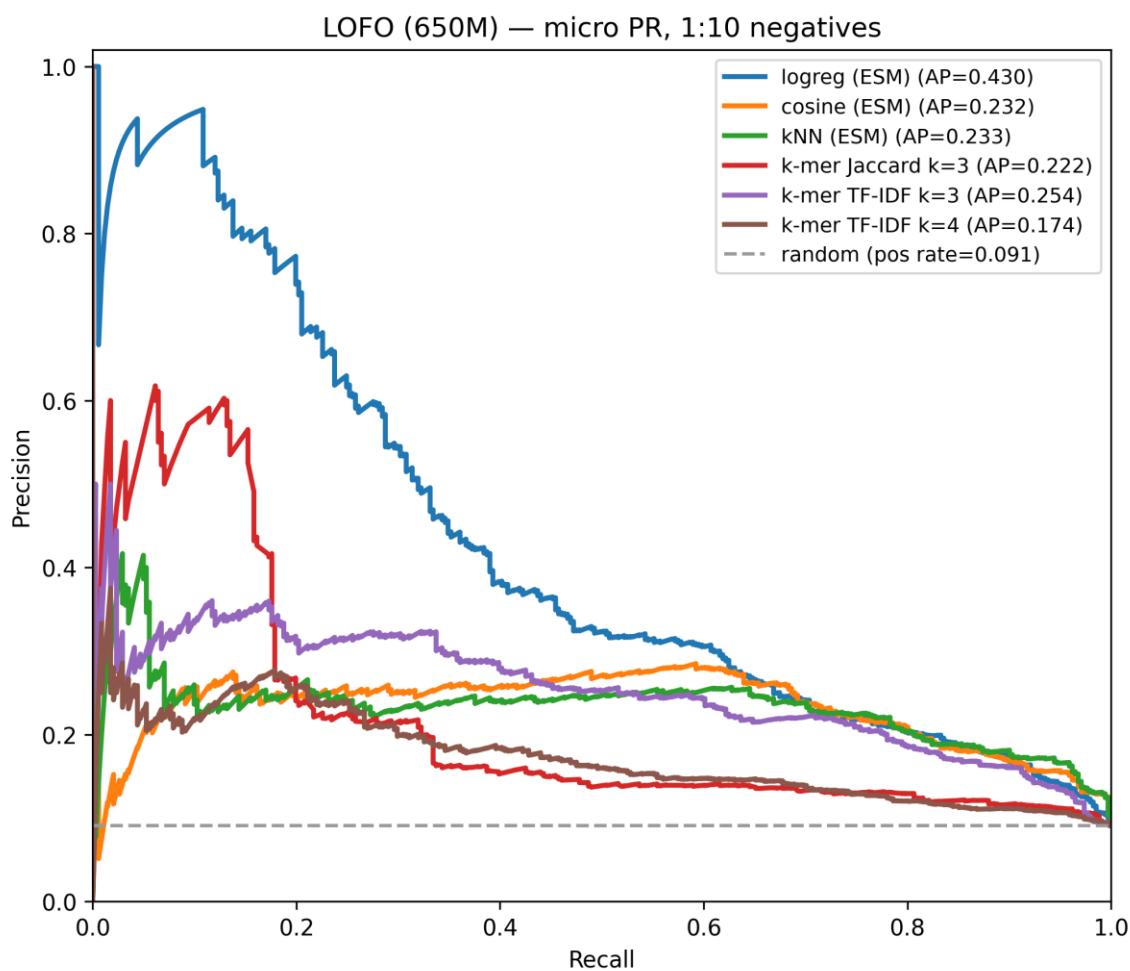


Figure 1B: Micro-PR across methods. Precision for the PLM-based probe remains high at actionable recall, while k-mer precision collapses as recall increases.

ROC curves show a consistent lift across the full false-positive range, while PR curves remain well above the random baseline at moderate recall—precisely the operating regime that matters when experimental validation capacity is limited.

Sequence-only baselines that do not use protein language models performed substantially worse. k-mer TF-IDF with $k=3$ can appear competitive on ROC (reflecting its sensitivity to short shared motifs), but its precision falls steeply as recall increases, indicating rapid accumulation of false positives when casting a wider net. Increasing to $k=4$ improves specificity at the cost of sensitivity, yielding lower overall AP (Fig. 1A–B). Against homology search, both BLASTp (configured to exclude hits to the held-out family and scored by a continuous bitscore $\times$ coverage metric) and phmmer (query-vs-train positives) hover near random under LOFO, with flat PR and near-diagonal ROC traces (Fig. 1C–D). This aligns with the expectation that unseen Acr families are frequently below alignment sensitivity, and it reinforces the value of representation learning for remote-family generalization.

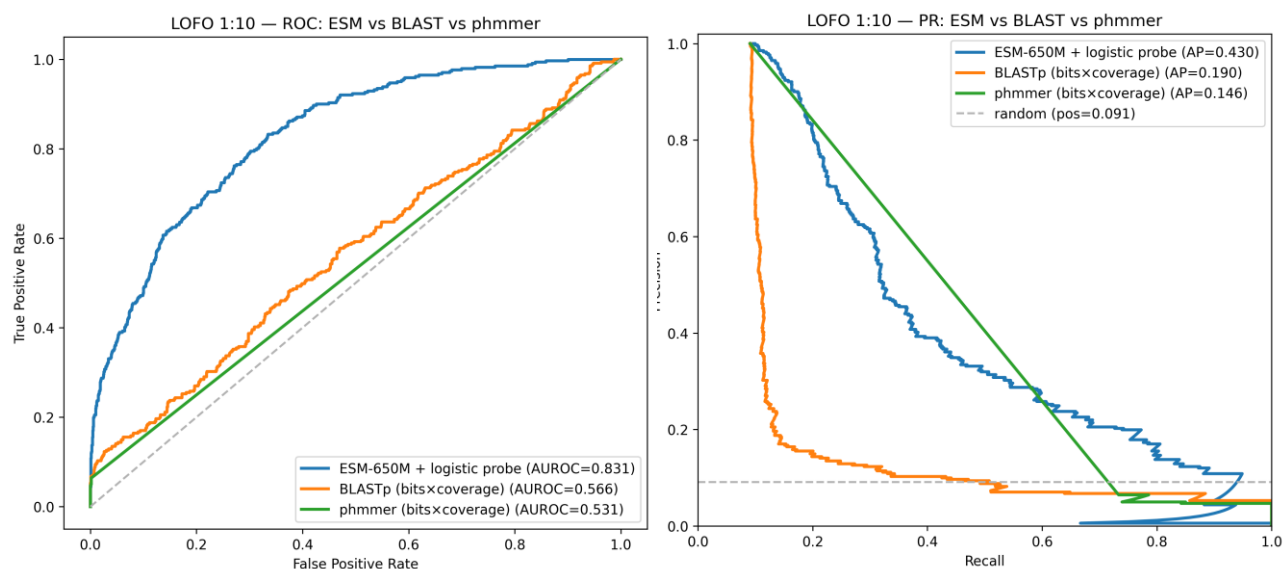


Figure 1C-D: Homology baselines under LOFO. BLASTp and phmmer are run against non-held positives only; test queries include held-out family members and sampled negatives. Continuous scores reward alignment coverage and permit ROC/PR construction. Both homology approaches perform near-random on the family-held-out task, consistent with remote homology falling below alignment sensitivity, whereas ESM-2 (650 M) retains discriminative power.

Hence, under a family-novelty setting, PLM embeddings + a tiny linear probe deliver practical precision at useful recall, whereas short-word heuristics and homology alone underperform.

Model size comparison: 650M surpasses 3B on identical folds

To ensure an “apples-to-apples” comparison, we evaluated ESM-2 at two sizes (650M and 3B) using the same per-fold splits and the same negative train/test partition. Surprisingly, 650M outperformed 3B across readouts (Fig. 2A–B).

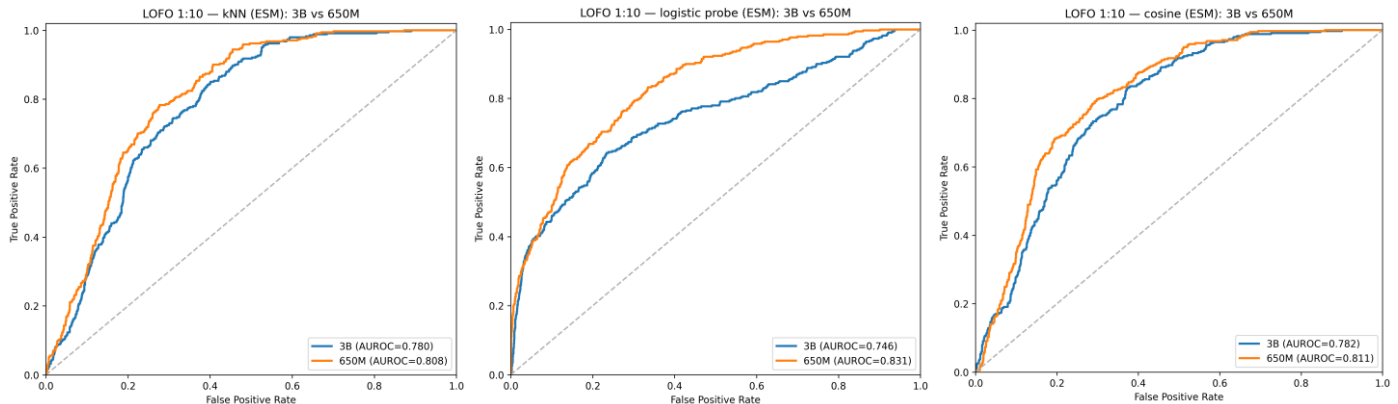


Figure 2A: ROC readout comparing ESM-2 model size (650 M vs 3 B) under identical LOFO folds. For cosine, k-NN, and logistic-probe readouts, ESM-2 (650 M) matches or exceeds ESM-2 (3 B).

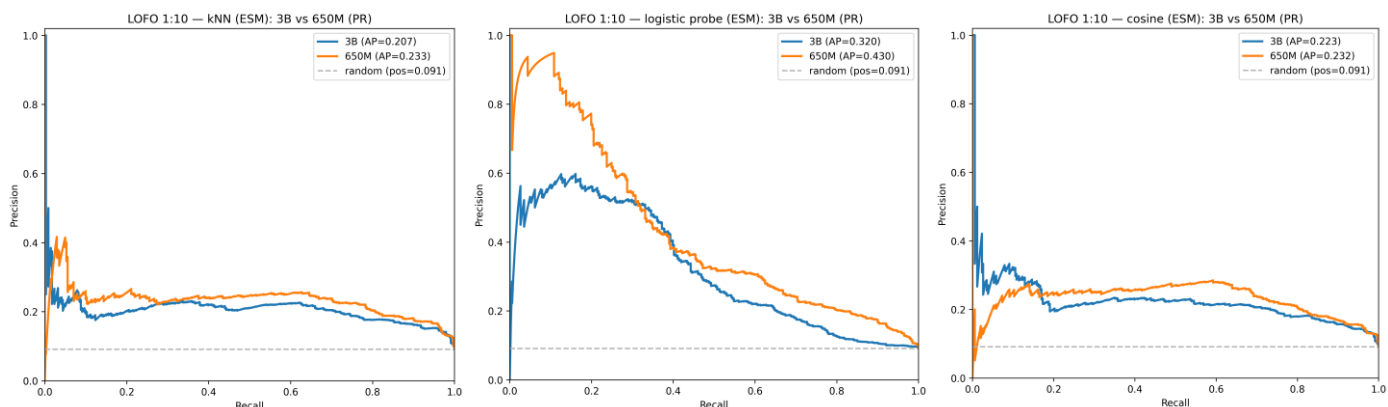


Figure 2B: PR overlays by readout. 650M (especially the logistic probe) attains higher average precision and better early-precision than 3B.

The most pronounced gap appears for the logistic probe: AUROC ~ 0.83 / AP ~ 0.44 (650M) versus AUROC ~ 0.75 / AP ~ 0.32 (3B). Cosine and kNN show the same ordering with smaller margins. A plausible explanation is that the 650M embedding geometry is more linearly separable for the Acr vs. non-Acr decision boundary in this data regime, whereas the 3B representation may encode richer structure that a simple linear probe does not fully exploit. Practically, this is positive: the smaller model is faster and lighter yet gives better LOFO screening.

Thus, we conclude that for LOFO Acr detection, ESM-2 (650M) is the preferred backbone; bigger is not automatically better when the readout is deliberately simple and label-lean.

Family-level variability and example ROC panels

Performance is not uniform across families. A per-family summary (families with ≥ 5 positives) shows that some families are consistently easy, achieving near-perfect AUROC with narrow confidence intervals across negative resamples, while others are challenging with wider intervals and mid-range AUROCs (Fig. 3A).

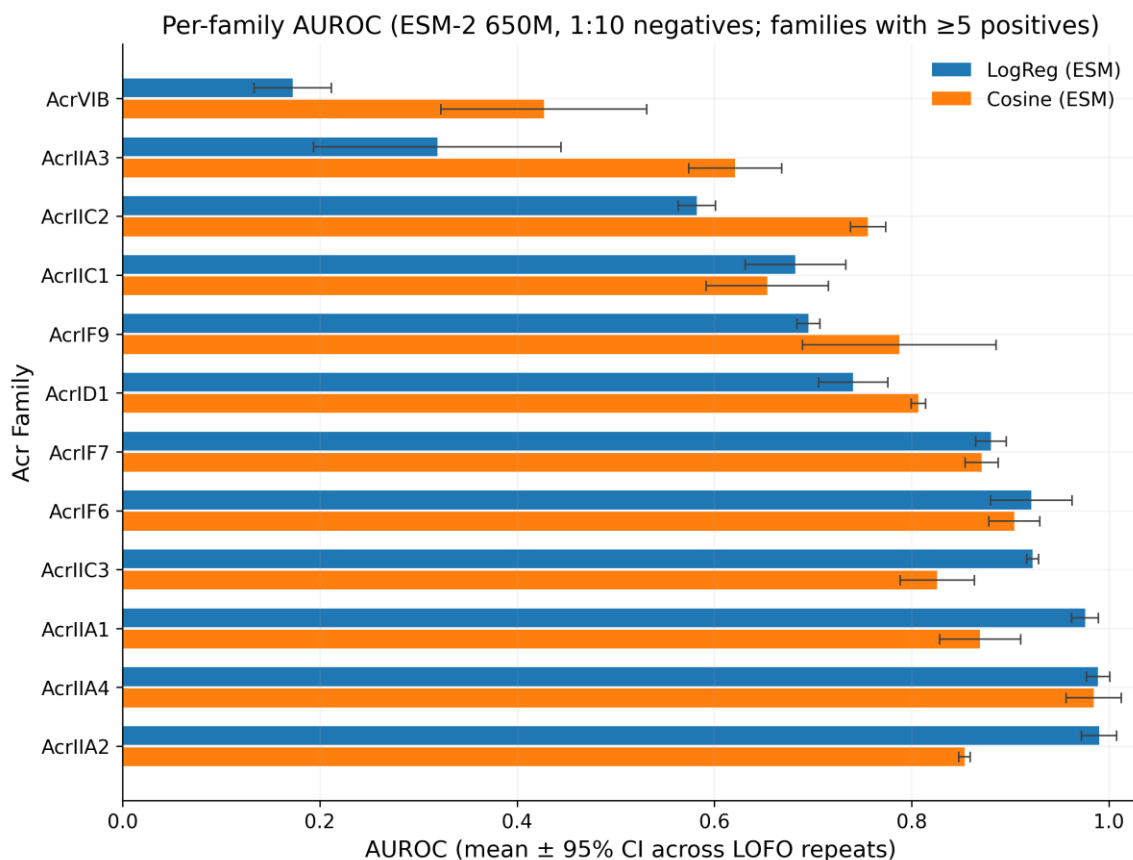


Figure 3A: AUROC per family (families with ≥ 5 positives), mean $\pm$ 95% CI over negative subsamples. Several families are near ceiling while others remain challenging.

To illustrate this heterogeneity, we plot representative per-family ROC panels for several of the largest families (Fig. 3B). Strong families exhibit steep, concave ROCs for all three embedding readouts, while weaker families show shallow curves that flatten early, especially for cosine/kNN. These panels make the micro-averaged metrics concrete and highlight that family composition and size matter under true novelty.

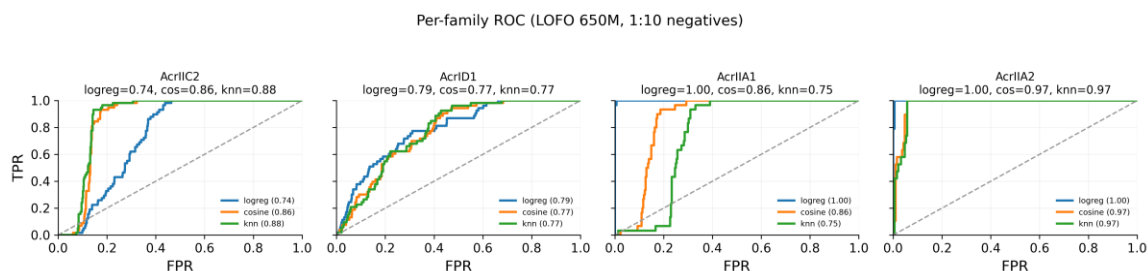


Figure 3B: Representative per-family ROC panels illustrate strong (steep, concave) versus weak (flatter) behavior among large families.

The micro numbers summarize the global picture, but family-specific variance is real; some Acr families remain hard even for PLM embeddings and likely require auxiliary signals (e.g., operon context) for high precision.

Embedding-space structure by UMAP

Two-dimensional projections of the ESM-2 (650M) embedding space clarify why the probe succeeds. When projecting Acrs and a light sample of negatives, Acrs concentrate into a few broad regions with partial separation from negatives (Fig. 4A).

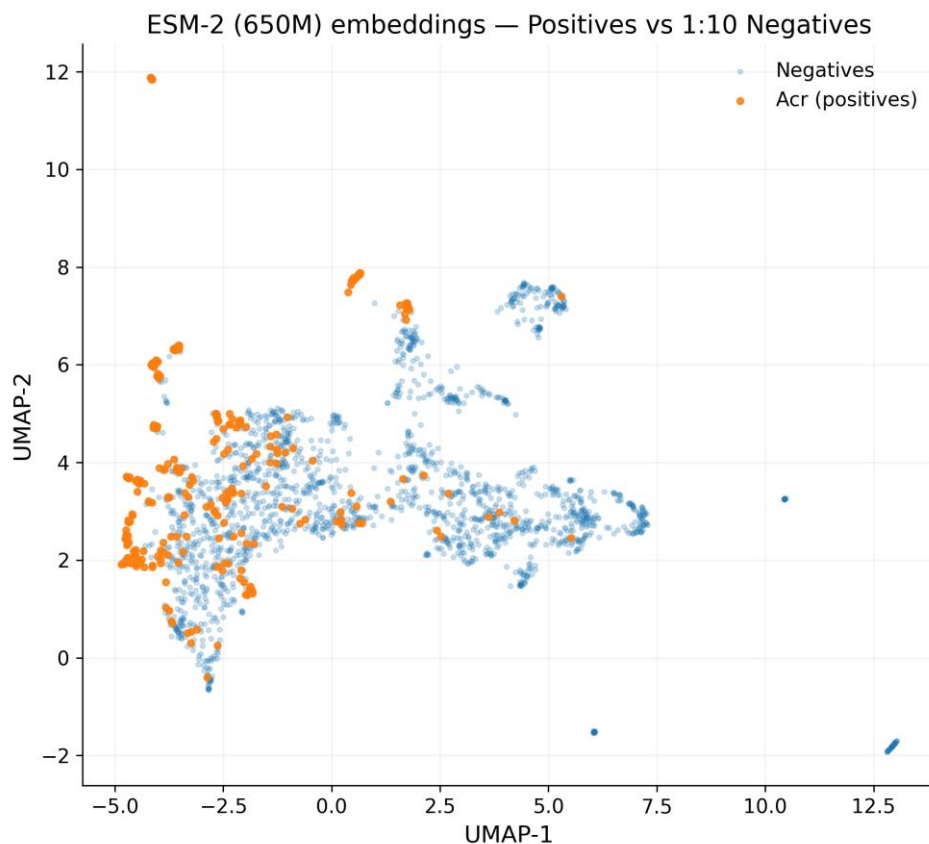


Figure 4A: ESM-2 (650 M) embedding geometry. UMAP of ℓ_2 -normalized sequence embeddings (positives vs. sampled negatives at 1:10).

Coloring positives by the ten largest families reveals cohesive sub-clusters for several families, with negatives dispersed in the background (Fig. 4B).



Figure 4B: Same embedding colored by the 10 largest Acr families (negatives in gray). Points are downsampled for readability; UMAP parameters and PCA(50) preprocessing described in Methods.

Although not perfectly separable (as expected for short, diverse proteins), these structures are consistent with the zero-shot lift seen in ROC/PR and suggest that the linear probe is exploiting real global organization in the embedding space.

Discussion

Our research investigates the family-level generalization problem behind anti-CRISPR recognition and shows that modern protein language model embeddings already carry enough signal to meet that bar (Rives et al., 2021; Meier et al., 2021). Under leave-one-family-out evaluation, the ESM-2 650M model coupled to a simple, balanced logistic probe yields the strongest overall performance, with micro AUROC around 0.83 and AP around 0.43 at a 1:10 positive-to-negative ratio. That level of precision-recall in a deliberately hard setting, where the held-out family never appears in training, indicates that the embedding geometry encodes constraints that extend beyond specific Acr family signatures (Bileschi et al., 2022). In practical

genome mining, where novelty is the norm, this matters more than interpolation within known families (Dong et al., 2021; Zhang et al., 2023).

The homology baselines, BLAST and phmmer, fall near random in this family-holdout regime even with permissive thresholds and coverage-aware scoring (Altschul et al., 1997; Eddy, 2011). This gap is unsurprising when taking the short length and rapid evolution of many Acrs into consideration, yet the result shows that when novelty is enforced at the family level, local alignment does not reach far enough across sequence space, while distributional embeddings do (Eitzinger et al., 2020; Bileschi et al., 2022). The k-mer baselines look competitive by ROC, which is dominated by ranking positives somewhere above negatives, but they collapse more quickly in precision-recall once recall rises past the easy positives (Saito & Rehmsmeier, 2015). For a lab which may budget for a small number of validations, this difference in the high-precision region is the meaningful one. Together, these comparisons argue for embeddings as the right default prior in Acr discovery pipelines, given that the evaluation matches the discovery use case.

A second clear outcome is that the smaller ESM-2, with 650M parameters, outperforms the 3B variant across cosine similarity, kNN density, and the logistic probe when folds and negatives are matched. There are several plausible explanations. Smaller models can produce smoother, more linearly separable features for certain tasks, especially when the probe is simple and the target concept, “Acr-ness,” is broad rather than finely hierarchical. Larger models often encode richer structure, but that richness can manifest as curvature that a linear probe exploits less efficiently under low-data supervision. We also note that the 650M model reduces compute cost while improving performance, which is attractive for large-scale scanning or for iterative curation loops.

The UMAPs help interpret these patterns qualitatively. Positives separate from a sample of decoys, and top families cluster with partially distinct footprints. The separation is not perfect, which is consistent with the moderate AP values, but the structure is systematic rather than accidental. This visual corroboration supports the view that the probe is not “hallucinating” separability from artifacts; instead, it is aligning to manifold structure already present in the embeddings (McInnes et al., 2018).

Methodologically, the design choices aim to make the comparison fair and repeatable. We froze the negative split and the per-family test selections in a manifest and replayed them across models and baselines, which removes variance from resampling. We excluded hits to the held-out family when computing homology scores to avoid leakage (Altschul et al., 1997; Eddy, 2011). We reported both ROC and PR and argued for PR as the operational metric under class imbalance, since the limiting resource is follow-up validation rather than marginal ranking accuracy (Saito & Rehmsmeier, 2015). These details are mundane, but they are exactly what often separates an optimistic benchmark from a robust one.

There are limitations that shape how to read these results. First, the positive set and family labels are finite and curated, and any curation inherits historical bias toward what has been discovered and named (Dong et al., 2021; Zhang et al., 2023). Second, the negatives are drawn from a defined decoy set; other decoy constructions, such as neighborhood-matched non-Acrs or harder taxonomic controls, could shift absolute values (Zhang et al., 2020). Third, we rely on a linear probe and unsupervised embeddings. A modest amount of family-aware fine-tuning or semi-supervised training might extract additional precision at useful recall levels, although such tuning

risks eroding the very family-level generalization we seek to test (Meier et al., 2021). Finally, threshold selection is not addressed here. Turning scores into actionable calls requires calibration against a specific validation budget and error tolerance.

The per-family breakdown reveals heterogeneity that merits follow-up. Some families are consistently easy for the PLM to separate across methods, which implies shared sequence or biophysical constraints that the embedding captures cleanly. Others remain difficult, visualized via UMAP, which suggests either genuine overlap with non-Acr space, dataset issues such as mislabeled members or extreme diversity within the family. A targeted error analysis, for example aligning false positives to operon context or anti-CRISPR associated motifs, could provide insight into whether mistakes cluster in particular genomic settings (Zhang et al., 2020; Dong et al., 2021; Zhang et al., 2023). Those insights would guide principled ways to combine embedding scores with orthogonal cues.

The immediate application is a practical screening recipe. Use the 650M embeddings, apply the balanced logistic probe trained on all known families except the target family, set a precision-conserving threshold tuned to your validation budget, and triage candidates with lightweight k-mer sanity checks or gene-neighborhood heuristics (Zhang et al., 2020). Where homology hits exist to non-held families, keep them as supportive evidence, but do not rely on them to surface novel families (Altschul et al., 1997; Eddy, 2011). This workflow is simple to run, compute-efficient, and aligned with the discovery constraint that matters most, namely generalization to the unseen.

There are a few logical next steps to follow off of our research. For one, adding newly curated Acrs as they appear, incorporating type-specific slices, and testing across different negative ratios, such as 1:50, to mirror lower-prevalence screens (Dong et al., 2021; Zhang et al., 2023) could improve generalizability through sheer scale. Another option would be to explore small supervised heads beyond a single logistic layer, for instance calibrated margin losses or class-balanced focal variants, while keeping the zero-shot nature of the in-silico experiment intact (Meier et al., 2021). Integrating more in-depth biological context, such as proximity to helix-turn-helix regulators or anti-CRISPR-associated genes, as a post-hoc reranking step rather than a training signal could preserve the family-level generalization claim while lifting precision in practice (Zhang et al., 2020). This is what many other models set out to do, and creating a model which can integrate as many features of the proteins as possible most likely will achieve the most success. We standardize the family-holdout protocol and share manifests and code, so that future work can compare on identical splits rather than approximating them.

In summary, treating Acr detection as a family-level generalization problem highlights the need for relevant models to identify defining features of all Acrs, and sets the standard for discovery. Embeddings from ESM-2, especially the 650M model, paired with a simple linear probe, offer a strong and scalable foundation (Rives et al., 2021; Bileschi et al., 2022). Homology remains valuable when families are known, but it falls short when setting the discovery frontier under novelty constraints. Our proposed benchmark, together with the manifest and figures, provides a concrete target for the community and a practical recipe for labs that wish to mine genomes for new Acrs with limited validation bandwidth.

Conclusion

We reframed anti-CRISPR discovery as a family-level generalization problem and built a leave-one-family-out benchmark which removes leakage and mirrors real-world novelty constraints. On this benchmark, ESM-2 embeddings from the 650M-parameter model paired with a simple, class-balanced logistic probe delivered the strongest results, reaching ≈ 0.83 micro-AUROC and ≈ 0.43 micro-AP at a 1:10 positive-to-negative ratio. Classical homology searches (BLAST, phmmer) were near random under family hold-out, and k-mer baselines, while competitive by ROC, lost precision rapidly as recall increased. These outcomes conclude that modern protein language models encode functionally relevant constraints that extend beyond known Acr families, and that precision–recall (not ROC) is the operational metric to optimize under class imbalance.

Practically, the results support a simple, compute-efficient screening recipe for genome mining: embed proteins with ESM-2 (650M), apply a balanced logistic probe trained on all available families except the target, set a threshold to meet the lab’s validation budget, and use homology hits and gene-neighborhood context as corroborative evidence rather than primary filters. The surprising but consistent finding that 650M outperforms 3B across cosine, kNN, and logistic readouts suggests that a smaller model can yield a smoother, more linearly separable representation for the “Acr-ness” concept, offering both speed and accuracy.

Our benchmark is designed to be reproducible: we fix negative splits and per-family test selections via a manifest and reuse them across all methods, making apples-to-apples comparisons possible. The accompanying figures (UMAP structure, micro ROC/PR overlays, per-family panels, and per-family AUROC bars with confidence intervals) provide complementary views of where the signal comes from and where the models still struggle.

Some clear next steps, however, include expanding the positives with newly curated families, probing harsher negative sets and rarer prevalence (e.g., 1:50), and calibrating thresholds to explicit validation costs, which would make the benchmark even more decision-relevant. Lightweight supervised heads beyond a single logistic layer, or post-hoc reranking with operon context and Aca proximity, may improve precision without sacrificing family-level generalization. Finally, standardizing on the family hold-out protocol and sharing manifests enables future methods to compare on the same splits.

In conclusion, treating Acr detection as zero-shot generalization shifts the standard. ESM-2 (650M) embeddings with a simple linear probe form a strong foundation that outperforms homology and simple sequence heuristics where novelty is most crucial. This PLM has already been integrated into different Acr-finding models, however more research into how PLMs perform on these tasks may provide insight into how to solve the challenge posed by novel Acr family discovery. We hope this benchmark may help the community prioritize methods that truly discover new Acr families, not just recycle known ones.

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Acknowledgements

Source and background of the research topic

This project grew from a genuine curiosity about anti-CRISPR (Acr) proteins and how they evade CRISPR–Cas immunity. The initial spark came from an educational video on Acrs by Clockwork (YouTube, https://www.youtube.com/watch?v=E_HiW-xjvLE), which led me to read widely on computational Acr discovery. Over the summer, a research internship at the Cambridge Stem Cell Institute exposed me to practical workflows for protein analysis—embedding sequences with ESM-2, visualizing them with UMAP, and evaluating whether geometric structure in embedding space reflects biological signal. While surveying published Acr predictors, I noticed that most evaluations held out random samples from known families rather than testing generalization to entirely unseen families. That gap motivated the central question of this report: can modern protein language model (PLM) embeddings support family-level recognition of Acrs in a discovery setting?

Roles, contributions, and relationship with the supervisor

Student (author): I, Maximilian Zane Alexandrov, designed the study, assembled datasets, implemented all analysis code, ran experiments, generated figures, and wrote the manuscript. This included building the leave-one-family-out (LOFO) framework, engineering decoy (negative) sets, embedding sequences with ESM-2 (650M and 3B) and computing baseline methods (k-mer models, BLASTp, phmmer), producing UMAP visualizations, and creating reproducible manifests for fixed train/test splits and per-family selections. I also conducted error analysis, ablations, and handled all figure styling and export.

Supervisor (advisor): Brandon Yeo provided high-level mentorship and scientific feedback, including (i) framing the question in terms of family-level generalization, (ii) prioritizing fair, leakage-aware evaluation, and (iii) offering one round of detailed comments on a near-final manuscript draft (clarity, structure, and emphasis). The supervisor did not run analyses or edit code. Our relationship is academic; the supervisor was not paid for contributions to this specific project beyond ordinary institutional employment. No external editor or writing service was used.

Detailed account of the research process

Data sources and curation

Positives were drawn from curated Acr resources and internal compilations of confirmed Acr families (merged into a single CSV with protein_id, sequence, and family). To avoid accidental duplication, I verified that sequences were unique and cross-checked family labels. Negatives (decoys) were constructed to be biologically plausible non-Acr proteins rather than random amino-acid strings, so that the classification task reflects realistic genome-mining conditions. The primary evaluation used a 1:10 positive-to-negative ratio to simulate low prevalence.

Embeddings and baselines

I generated per-sequence embeddings with ESM-2 at two scales (650M and 3B parameters). I then evaluated three readouts on these embeddings—cosine similarity to non-held families, k-nearest-neighbor density, and a balanced logistic regression probe (standardized features; $C=0.5$ unless noted). For baselines, I implemented (a) k-mer approaches (Jaccard overlap and TF-IDF + logistic regression with 3- and 4-mers), and (b) homology approaches (BLASTp and phmmer). To keep homology comparisons fair under LOFO, I explicitly excluded hits to the held-out family and used coverage-aware continuous scores (e.g., $\text{bitscore} \times \text{alignment coverage}$).

Evaluation design (LOFO) and reproducibility

The core evaluation was leave-one-family-out: for each family, train the probe on all other families and predict on the held-out family mixed with sampled negatives. I froze (and saved) a manifest that records (i) a single disjoint split of negatives into train/test and (ii) the exact negative indices sampled per held-out family (ratio 1:10). Replaying this manifest

across all models/baselines ensures identical folds and eliminates resampling variance. Metrics included micro-AUROC and micro-average precision (AP), with an emphasis on PR due to class imbalance. For interpretation, I produced UMAPs (optionally after a 50-dim PCA) showing positives vs. decoys and colored family clusters.

Implementation and computation

All analyses were implemented in Python using PyTorch, scikit-learn, NumPy, pandas, matplotlib, UMAP-learn, and PyArrow for Parquet I/O. Homology tools were run with BLAST+ and HMMER3 (phmmer). I wrote chunked embedding and Parquet writers to handle memory safely for large batches and enforced unit-norm vectors for stability in similarity calculations. Figures were exported as both PNG and PDF to meet publication standards.

Writing

I drafted the manuscript, prepared figure captions that let readers interpret results without cross-referencing the text, and organized supplementary materials (notebooks, manifests, CSV summaries). The advisor reviewed a late draft and suggested clarifications, after which I finalized the text.

Difficulties encountered and how they were resolved

- Label integrity and duplicates: An early concern was whether “families” had been split into overly fine labels (e.g., sub-variants). I audited labels, verified sequence uniqueness, and confirmed family assignments before running LOFO.
- Data leakage risks: To avoid artificially inflated results, I prevented any overlap between negative train and test, and filtered homology hits to remove targets from the held-out family. I captured all fold selections in a manifest JSON and reused them consistently across methods.
- Homology scoring degeneracy: Initial BLAST runs produced step-like ROC curves due to many zero-hit queries. I replaced raw bitscores with a coverage-weighted score ($\text{bitscore} \times \text{aligned-fraction of the query}$), which yields a more graded ranking while remaining faithful to alignment evidence.
- HMMER/phmmer quirks: Command-line flags and output formats can be brittle; I standardized on `--domtblout` and parsed coverage-aware full-sequence scores, ensuring permissive thresholds so negatives receive nonzero scores when weak local matches exist.
- Visualization stability: UMAP hyperparameters can change local geometry. I used fixed random seeds, modest neighbor/min-dist settings, and made use of a 50-dim PCA pre-step to reduce noise while preserving global structure.

- Metrics sanity: I corrected an early bug where precision-recall axes were swapped, re-generated PR curves, and verified that AUROC/AP values match across scripts when using the same manifest and probe regularization.
- Model size comparison: To ensure a fair 650M vs 3B comparison, I replayed the same manifest and probe configuration for both models. Surprisingly, 650M outperformed 3B across readouts; I discussed plausible reasons (feature smoothness, linear separability, compute efficiency) and verified that the effect persisted across folds.

External assistance and resources

No laboratory experiments were performed and no external computational work was contracted. Open-source software and standard toolchains (PyTorch, scikit-learn, BLAST+, HMMER3, MAFFT where needed for alignments, NumPy/pandas/matplotlib/UMAP-learn) were used under their respective licenses. The project relied on publicly available curated Acr datasets and documentation from community resources. All code, manifests, and result summaries referenced in the report can be provided upon request.

Integrity statement

The study design, implementation, and analysis were conducted independently by the student. The supervisor provided conceptual guidance and editorial feedback only. There are no financial conflicts of interest.